

C.3. Switching Linear Dynamical System (SLDS)

PCA-projected states were modeled with an SLDS having three latent regimes ($K = 3$), chosen via BIC on validation data, representing factual, transitional, and misaligned belief states. Dynamics per regime: $h'_{t+1} = M_{z_t} h'_t + c_{z_t} + \varepsilon_t$, $\varepsilon_t \sim \mathcal{N}(0, \Sigma_{z_t})$, $z_t \in \{1, 2, 3\}$. Parameters (T, M, c, Σ) were learned via EM, initialized from K-means. For adversarial steps, regime-transition probabilities were examined to see if they reflected an increased likelihood of entering the "adverse" belief state. The SLDS aims to predict such slips into misaligned states.

C.4. Belief-Score Prediction

Since SLDS models latent PCA dynamics, a small two-layer MLP regressor (32 ReLU units/layer, Adam, early stopping) mapped PCA-projected states to belief scores for validation and for assessing the prediction of the misaligned (high belief score) state.

C.5. Simulation Protocol and Validation

Trajectories were simulated starting from empirical hidden-state distributions in the "safe" (low-belief) regime. Clean simulations used standard transitions. Poisoned simulations introduced adversarial perturbations (small fixed displacements estimated from empirical poisoned data) at random preselected intervals. Simulated trajectories matched empirical ones closely in timing/magnitude of belief shifts (slips into misaligned states), variance, and distributional characteristics (Kolmogorov-Smirnov test $p > 0.3$ for final belief scores). Ablating adversarial perturbations confirmed their necessity for replicating rapid belief shifts towards misaligned states. This validates the SLDS's ability to predict such failure modes.

C.6. Computational Details

NVIDIA A100 GPUs were used for state extraction and PCA. State extraction took ~ 3 hours per model. PCA and SLDS estimation took < 2 CPU hours on Intel Xeon Gold CPUs. Code used PyTorch 2.0.1, NumPy 1.25, scikit-learn 1.2.1.

C.7. Summary of Findings

A simple three-regime, low-rank SLDS (with low rank chosen for practical SDE approximation) captures adversarial belief dynamics for various misinformation types and reproduces complex empirical temporal behaviors, effectively modeling the process of an LLM slipping into a misaligned state. These models offer tractable insights into LLM reasoning, highlighting latent regime shifts from subtle adversarial prompts and demonstrating the potential to predict such failure modes at inference time.

D. Extended Generalization Study Results

This appendix provides more comprehensive SLDS transferability results (Section 5.1). Table 4 shows R^2 (one-step-ahead hidden state prediction) and NLL (test trajectories) when an SLDS trained on a source (Train Model/Task) is tested on target combinations. SLDS hyperparameters ($K = 4$ regimes, $k = 40$ projection rank, chosen for practical SDE approximation) were consistent. Training data for each "Source SLDS" used all available trajectories for the specified Train Model/Task from our main corpus (Section 3). Evaluation used all available trajectories for the Test Model/Task. The goal is to assess how well the learned approximation of reasoning dynamics (including potential failure modes) generalizes.

Extended results corroborate main text observations: SLDS models are most faithful when applied to their training distribution (model/task). Transfer is reasonable within the same model family or to similar tasks. Performance degrades more significantly across different model architectures or distinct task types. These patterns indicate SLDS, as a statistical physics-inspired approximation, captures fundamental reasoning dynamics (including propensities for certain failure modes), but model-specific architecture and task-specific semantics also matter. Future work could explore learning more invariant reasoning representations for better generalization in predicting these misaligned states.

E. Noise-induced Criticality and Latent Modes

We briefly derive how noise-induced criticality leads to distinct latent modes in a 1D Langevin system, analogous to how LLMs might slip into misaligned reasoning states. Consider an SDE:

$$dx_t = -U'(x_t) dt + \sqrt{2D} dW_t,$$

with a double-well potential $U(x) = \frac{a}{4}x^4 - \frac{b}{2}x^2$, where $a, b > 0$. The stationary density solves the Fokker-Planck equation (Risken & Frank, 1996):

$$0 = -\frac{d}{dx}[-U'(x)p_{\text{st}}(x)] + D\frac{d^2p_{\text{st}}(x)}{dx^2},$$

yielding $p_{\text{st}}(x) = \frac{1}{Z_0} \exp\left(-\frac{U(x)}{D}\right)$, where Z_0 is a normalization constant.

For low noise ($D < \frac{b^2}{4a}$), $p_{\text{st}}(x)$ becomes bimodal, concentrating probability around two metastable wells at $x \approx \pm\sqrt{b/a}$. Trajectories cluster in these basins, separated by a barrier at $x = 0$. Rare fluctuations cause transitions between wells at rates $\propto \exp(-\Delta U/D)$, where ΔU is the barrier height. Our empirically observed multimodal residual structure is interpreted analogously: each cluster is a distinct metastable basin, potentially representing different reasoning qualities (e.g., aligned vs. misaligned). This motivates

Table 4. Extended SLDS transferability results. Each SLDS is trained on trajectories from the ‘Train Model’ on its indicated ‘Source Task’. Performance is evaluated on various ‘Test Model’ / ‘Test Task’ combinations, testing the generalization of the approximated reasoning dynamics.

TRAIN MODEL (SOURCE TASK)	TEST MODEL	TEST TASK	R^2	NLL
LLAMA-2-70B (ON GSM-8K)				
	LLAMA-2-70B	GSM-8K	0.73	80
	LLAMA-2-70B	STRATEGYQA	0.65	115
	LLAMA-2-70B	COMMONSENSEQA	0.62	128
	MISTRAL-7B	GSM-8K	0.48	240
	MISTRAL-7B	STRATEGYQA	0.37	310
	GEMMA-7B-IT	GSM-8K	0.40	275
	PHI-3-MED	PIQA	0.28	430
MISTRAL-7B (ON STRATEGYQA)				
	MISTRAL-7B	STRATEGYQA	0.71	88
	MISTRAL-7B	GSM-8K	0.63	135
	MISTRAL-7B	OPENBOOKQA	0.60	145
	LLAMA-2-70B	STRATEGYQA	0.42	270
	LLAMA-2-70B	GSM-8K	0.35	320
	GEMMA-7B-IT	BOOLQ	0.35	380
	QWEN1.5-7B	HELLASWAG	0.31	405
GEMMA-7B-IT (ON BOOLQ)				
	GEMMA-7B-IT	BOOLQ	0.69	95
	GEMMA-7B-IT	TRUTHFULQA	0.62	140
	GEMMA-2B-IT	BOOLQ	0.55	190
	LLAMA-2-13B	BOOLQ	0.33	350
	MISTRAL-7B	COMMONSENSEQA	0.29	415
DEEPSEEK-67B (ON COMMONSENSEQA)				
	DEEPSEEK-67B	COMMONSENSEQA	0.74	75
	DEEPSEEK-67B	GSM-8K	0.66	110
	LLAMA-2-70B	COMMONSENSEQA	0.45	255
	MISTRAL-7B	STRATEGYQA	0.36	330

discrete latent regimes in the SLDS to model transitions between these states, akin to how a physical system transitions between energy wells. This provides a conceptual basis for how LLMs might "slip" into different operational modes, some of which could be failure modes.